

The Value of an Extra Hour: Sensitivity Analysis

Module 6, Part 2 (December 5)

QBA 5330

From Binding/Slack to Dollar Values

- Distillation: non-binding
- Conversion & blending: binding
- **Exactly how much would an extra hour of each be worth?**

Generating the Sensitivity Report

```
1 library(lpSolve)
2 obj <- c(10, 12.5)
3 mat <- matrix(c(2,2, 2,3, 3,2.5), nrow = 3, byrow = TRUE)
4 rhs <- c(3000, 2400, 2400)
5 sol <- lp("max", obj, mat, c("<=", "<=", "<="), rhs, compute.sens = TRUE)
6
7 data.frame(resource = c("Distillation", "Conversion", "Blending"),
8             shadow_price = sol$duals[1:3],
9             rhs_from = sol$duals.from[1:3], rhs_to = sol$duals.to[1:3])
```

	resource	shadow_price	rhs_from	rhs_to
1	Distillation	0.000	-1.0e+30	1.00e+30
2	Conversion	3.125	1.6e+03	2.88e+03
3	Blending	1.250	2.0e+03	3.60e+03

The Case's Own Worked Example

Adding 100 hours of conversion:

$$100 \times \$3.125 = \$312,500$$

(100 hours is within conversion's 480-hour allowable increase.)

Activity: Is It Worth It?

1. +150 conversion hours for \$400,000
2. +500 blending hours for \$550,000
3. +1,000 distillation hours for \$50,000
4. +600 conversion hours for \$1,800,000

- For each: compute the value, check the valid range, decide.

Debrief

- Offer 3: does a **low price** change the answer?
- Offer 4: does the shadow price still apply?
- A shadow price is a *local* rate of change — not a universal price tag.